

CHALLENGE 01

Development Feasibility & Pro Forma Navigator

Build an AI-driven tool that generates a Development Ease Score for potential housing sites.

HOW TO USE THESE BRIEFS

These briefs consolidate the original challenge set and the additional ideas developed in review. Each is broad enough to support multiple approaches while still naming a concrete user, decision, and demo.

Teams should build a working prototype, use real regional data where possible, explain assumptions and uncertainty, and demonstrate the product with at least one

Pittsburgh or Allegheny County use case.

PROBLEM STATEMENT

The region's older housing stock and constrained development environment create significant friction for new housing production, especially affordable and starter homes. Developers and planners struggle to understand how zoning regulations, infrastructure limitations, and environmental constraints interact to affect project feasibility. Without a clear way to quantify these barriers, development pipelines remain slow, costly, and unpredictable.

This challenge seeks an AI-driven tool that generates a Development Ease Score for potential sites, highlighting regulatory bottlenecks, required variances, and infrastructure gaps so cities and developers can prioritize fast-track sites and identify areas needing policy or resource intervention.

SUCCESS LOOKS LIKE

A user can enter a parcel ID or compare multiple parcels and receive a source-grounded Development Ease Score, a plain-language explanation of the biggest barriers, and clear flags for zoning, environmental, infrastructure, or policy issues that require further review.

Owner / input

City, County, and PHFA with academic and mentor support.

Personas

- **Municipal Planner:** Needs tools to quickly assess feasible sites and identify regulatory bottlenecks.
- **Small/Mid-Size Developer:** Wants to reduce project risk and quickly evaluate site feasibility.
- **Housing Nonprofit/CDC:** Needs clarity on where affordable housing projects are most viable.
- **Policy Analyst:** Requires data to understand where zoning or infrastructure changes are most impactful.

Use cases

- Developer uploads a parcel ID and receives a Development Ease Score and insights on bottlenecks.
- Planner compares multiple parcels to identify sites best suited for starter home development.
- City identifies low-scoring sites and evaluates whether zoning reform or infrastructure upgrades are needed.
- Agencies prioritize investments based on score clusters across neighborhoods.

Publicly available data sources

- Allegheny County Real Estate Portal
- City of Pittsburgh Zoning Code & Zoning Map
- PA DEP eMapPA environmental datasets
- Pittsburgh Department of City Planning GIS

READY TO BUILD?

Use this brief to test a real parcel and housing concept.

The strongest submissions will make site feasibility easier to compare while showing the sources, assumptions, and human-review points behind each score.

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AI Horizons Pittsburgh
AI for Housing Hackathon

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